

TNA Corollary: Intuition of Post-Mortem Continuity (IPC)

A structural property of the human cognitive system

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Abstract

This paper introduces the *IPC Corollary (Intuition of Post-Mortem Continuity)* within the framework of the Theorem of Axiomatic Necessity (TNA). It argues that the human tendency to assume some form of continuity after death is not accidental nor merely cultural, but emerges from the structural constraints imposed by how the human mind represents the continuity of the Self under extreme uncertainty.

The corollary does **not** claim that post-mortem survival exists in reality. It explains why the human mind, even with insufficient empirical evidence, *almost surely converges* to the hypothesis of continuity.

1. Introduction

Since the Upper Paleolithic, humans have performed funerary rites, placed symbolic objects with the dead, and behaved as if existence continues beyond death. The universality of this phenomenon suggests that it is not an arbitrary cultural artifact but a deep cognitive feature.

This work formulates that feature as a corollary in the TNA framework:

The intuition of post-mortem continuity is a necessary consequence of how the mind models the continuity of the Self.

2. Theoretical Framework: Theorem of Axiomatic Necessity (TNA)

TNA begins with a minimal set of axioms about cognitive function:

1. **Axiom of Self-Continuity:**

The cognitive system represents personal identity as a temporally continuous trajectory.

2. **Axiom of Non-Representability of Non-Existence:**

The system cannot represent itself in a state in which it does not exist.

3. **Axiom of Structural Bayesian Updating:**

The mind updates internal probabilities under uncertainty through a process analogous to Bayesian updating, even with incomplete evidence.

Under these axioms, certain conclusions about the human experience of existential limits (death) follow necessarily.

3. Definition of the IPC Corollary

Let A be a human cognitive agent. Define:

$$\text{IPC}(A) = \Pr[\text{post-mortem continuity} \mid \text{cognitive structure of } A].$$

The **IPC Corollary** states:

Even under maximal uncertainty and insufficient evidence, the internal probability assigned to post-mortem continuity converges to 1 in the limit of cognitive updating.

Formally:

$$\lim_{n \rightarrow \infty} \Pr_n[\text{post-mortem continuity}] = 1.$$

4. Core Idea: A Representability Constraint

The key insight is that the opposite hypothesis:

- **H₀:** “there is no continuity”

cannot be coherently represented by a system whose sense of self requires continuity.

To model “my non-existence” forces a representational collapse: there is no possible internal state that can host that idea without contradicting the system’s identity architecture.

Thus:

$$\Pr[\text{non-continuity}] \approx 0.$$

And, by complementarity:

$$\Pr[\text{continuity}] \approx 1.$$

This is not cultural bias; it is a structural limitation.

What the mind cannot represent cannot accumulate probability.

5. Proof (Simplified Formal Version)

Define two hypotheses:

- H_1 : post-mortem continuity exists.
- H_0 : post-mortem continuity does not exist.

Assume an updating process:

$$\Pr_{n+1}(H_i) = f(\Pr_n(H_i), E_n),$$

where E_n is the evidence at step n .

By the Axiom of Non-Representability:

1. H_0 cannot be represented without contradiction.
2. Therefore any evidence E_n that would increase $\Pr(H_0)$ is discarded or reinterpreted.
3. Thus there exists an $\epsilon > 0$ such that:

$$\Pr_n(H_0) \leq \epsilon \quad \text{for all } n.$$

Taking the limit:

$$\lim_{n \rightarrow \infty} \Pr_n(H_0) = 0.$$

Finally:

$$\lim_{n \rightarrow \infty} \Pr_n(H_1) = 1.$$

6. Cross-Cultural Implications

The corollary explains:

- the universality of funerary behavior,
- the persistence of afterlife concepts across all cultures,
- the cognitive resistance to the notion of annihilation,
- the spontaneous emergence of ideas of “presence,” “continuity,” or “soul” even in atheists.

The phenomenon does not require cultural transmission; it is **pre-cultural**, produced by the architecture of the Self.

7. Discussion

The IPC Corollary does not attempt to solve the metaphysical question of post-mortem existence. Instead, it explains mathematically why the human mind:

- converges toward continuity,
- rejects annihilation,
- and builds mythologies, religions, and rituals at the existential boundary.

The phenomenon does not require belief; it arises from structural constraints.

8. Conclusion

The human intuition of continuity after death is neither an evolutionary accident nor a cultural superstition. It is a limit behavior of the cognitive system, derived from its structural inability to model its own non-existence.

The IPC Corollary provides a formal and universal explanation for one of humanity’s most persistent psychological intuitions.

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This work is part of the TNA series:

- TNA original → 10.5281/zenodo.17684115
- Bayesian ranking → 10.5281/zenodo.17744381
- Non-smoothness corollary → 10.5281/zenodo.17771689